CONTEXT BASED SHORT STORY GENERATION BY FINE TUNING LARGE LANGUAGE MODELS (LLMS)

~ Harsh Mamania

PROBLEM STATEMENT

- Interactive creative storytelling with generative language models has emerged in recent years as a novel and engaging means of gaming
- This project addresses the challenge of creating an AI-powered interactive storytelling system that can generate contextually appropriate narrative continuations.
- The primary goal is to fine-tune pre-trained language models to understand both the broader narrative context and specific user inputs to produce coherent, engaging, and contextually relevant story continuations.

MOTIVATION

- **AI Dungeon** - single-player text adventure game, provides story telling and textural adventure through generative fantasy responses
- *"Dungeons and Dragons (DnD) as a Dialog Challenge for Artificial Intelligence"* - Chris Callison, Gaurav Singh, Lara Martin et.al
- *"Context-Tuning: Learning Contextualized Prompts for Natural Language Generation"* – Tainyi Tang, Junyi Li, Wayne Xin Zhao.

DATASET – DEFINING CONTEXT IN SHORT STORIES

Created from processing 1002 Short-stories from Project Gutenberg (Kaggle Link)

- Originates from portal of 70,000 e-books containing short stories (Project Gutenberg)
- Summarization of excerpts from 1002 Short Stories

- Demonstrates concept of maintaining context in a story to LLM
- Size – **7748 Rows** in CSV format. Structure –> **Context – Input – Output**

Context

The winds blow Cappen off his feet as the ship drifts into the storm. Svearek tries to steer the ship closer to a large island in the storm, but it is too deep for him to row very well. He falls overboard and lands on a nearby lake. The other sailors help him onto a small boat that has been lowered into the sea by a strong wind….

Input

It turns out that the northmen fear

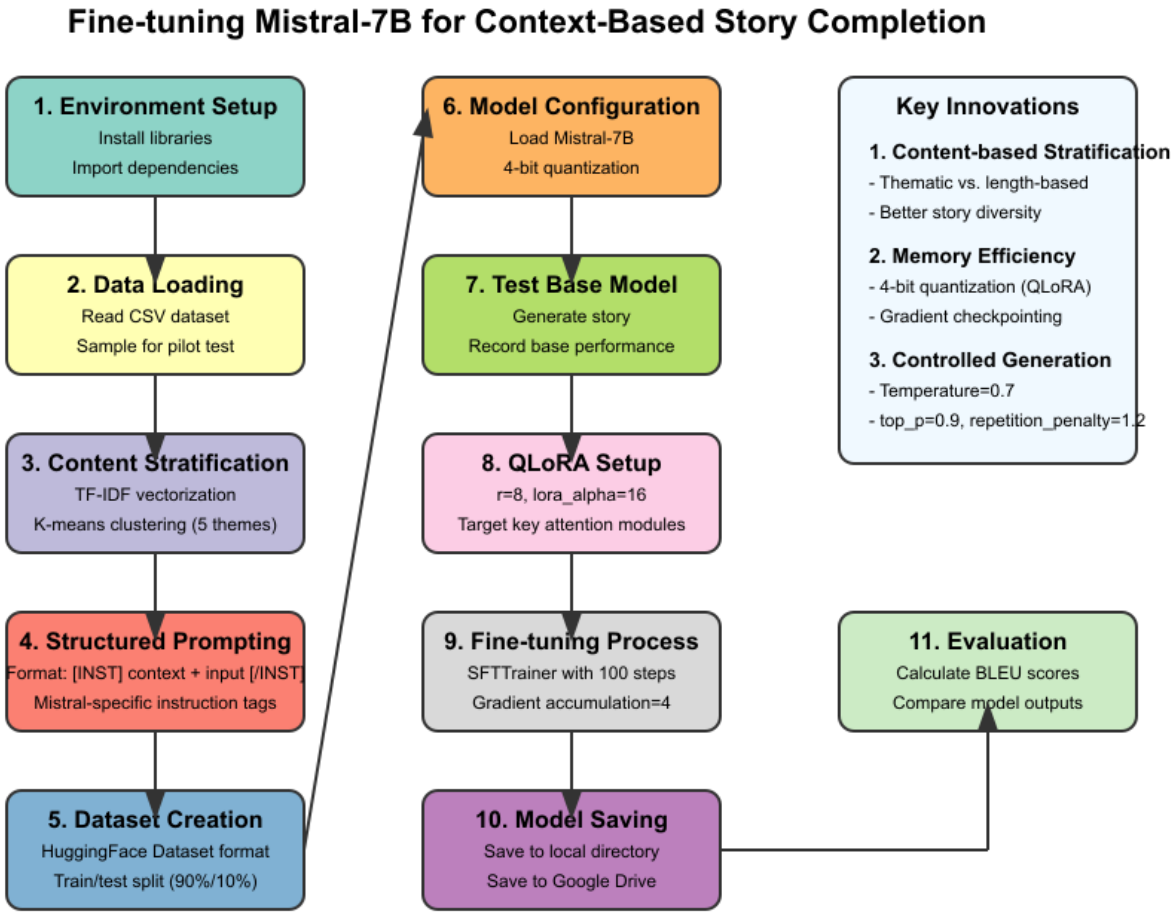
Output

It turns out that the northmen fear the southmen because they do not believe in ghosts or supernatural beings. So they send them back to their hiding place. As they get closer to the island, they come upon a cave filled with tunnels leading up to a massive rock wall. Inside, they can see the skulls of several men and hear the laughter of the troll-wife.

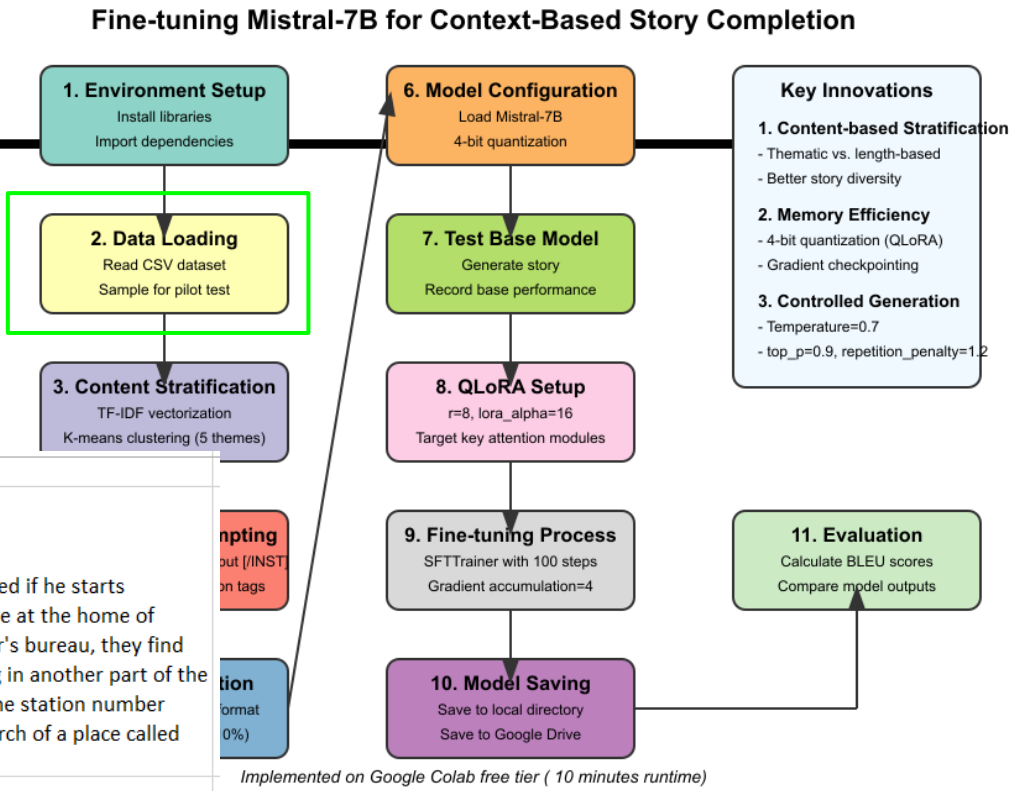
DATASET

	context	input	output
1	S. copyright on this publication was renewed.] Jeff needed a job and this man had a job to offer--one where giant economy-size trouble had labels like fakemake, bumsy and peekage! At first, Jeff scarcely notices the bold-looking man at the next table. Ann and Ann tease him about his "time travel," but he tells them that it's not as fun as Disneyland or Disney because they don't have secrets and no money.	Ann says she'll kick him cross-eyed if	Ann says she'll kick him cross-eyed if he starts anything. When they finally arrive at the home of Peter Powers, an agent of Snader's bureau, they find out that Mr. Snader is now living in another part of the United States. He has changed the station number from 725 to 724 . They go in search of a place called Green Thru-Way.
2	The winds blow Cappen off his feet as the ship drifts into the storm. Svearek tries to steer the ship closer to a large island in the storm, but it is too deep for him to row very well. He falls overboard and lands on a nearby lake. The other sailors help him onto a small boat that has been lowered into the sea by a strong wind. When they are able to board the boat, they find themselves alone on an uninhabited island with no one else to talk to. They all laugh at the idea of fighting over their lost love. Cappens first thinks of Ydris in Seilles, but then he realizes that	It turns out that the northmen fear	It turns out that the northmen fear the southmen because they do not believe in ghosts or supernatural beings. So they send them back to their hiding place. As they get closer to the island, they come upon a cave filled with tunnels leading up to a massive rock wall. Inside, they can see the skulls of several men and hear the laughter of the troll-wife.

APPROACH #1 - FINE-TUNING WITH MISTRAL7B-INSTRUCT



Implemented on Google Colab free tier (10 minutes runtime)



	context	input	output
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- The ***context*** is the context of the story,
- and for every incomplete **input** of about 5 to 6 words,
- there is an **output** of 3 or 4 lines *according to the context*.

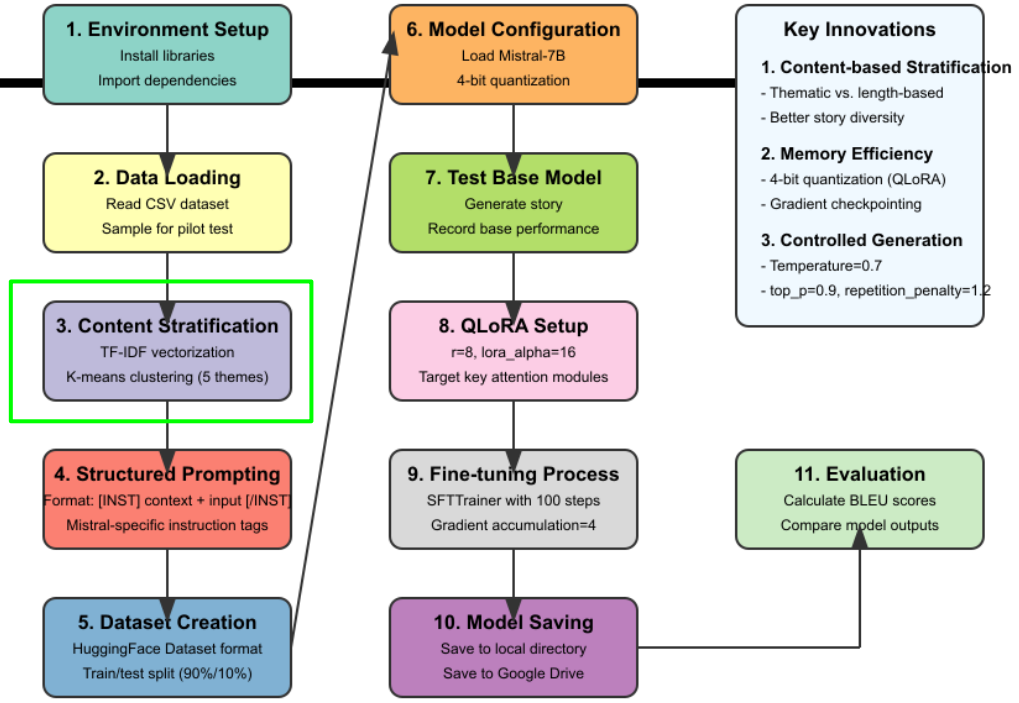
```
Implementing content-based stratification...
Using 5 thematic clusters

Top terms per cluster:
Cluster 0: time, one, tells, man, asks, explains, way, two, like, world
Cluster 1: uncover, extensive, evidence, research, copyright, renewed, publication, science, etext, produced
Cluster 2: old, man, mother, father, years, body, called, house, god, one
Cluster 3: mrs, narrator, story, says, tells, also, going, man, william, young
Cluster 4: stories, company, new, volume, author, story, short, english, tales, novels

Distribution of samples by theme cluster:
theme_cluster
0    263
2     82
3     65
4     60
1     30
Name: count, dtype: int64

Sample size: 148
Distribution in sample:
theme_cluster
0    78
2    24
3    19
4    18
1     9
Name: count, dtype: int64
```

Fine-tuning Mistral-7B for Context-Based Story Completion



Implemented on Google Colab free tier (10 minutes runtime)

	context	input	output
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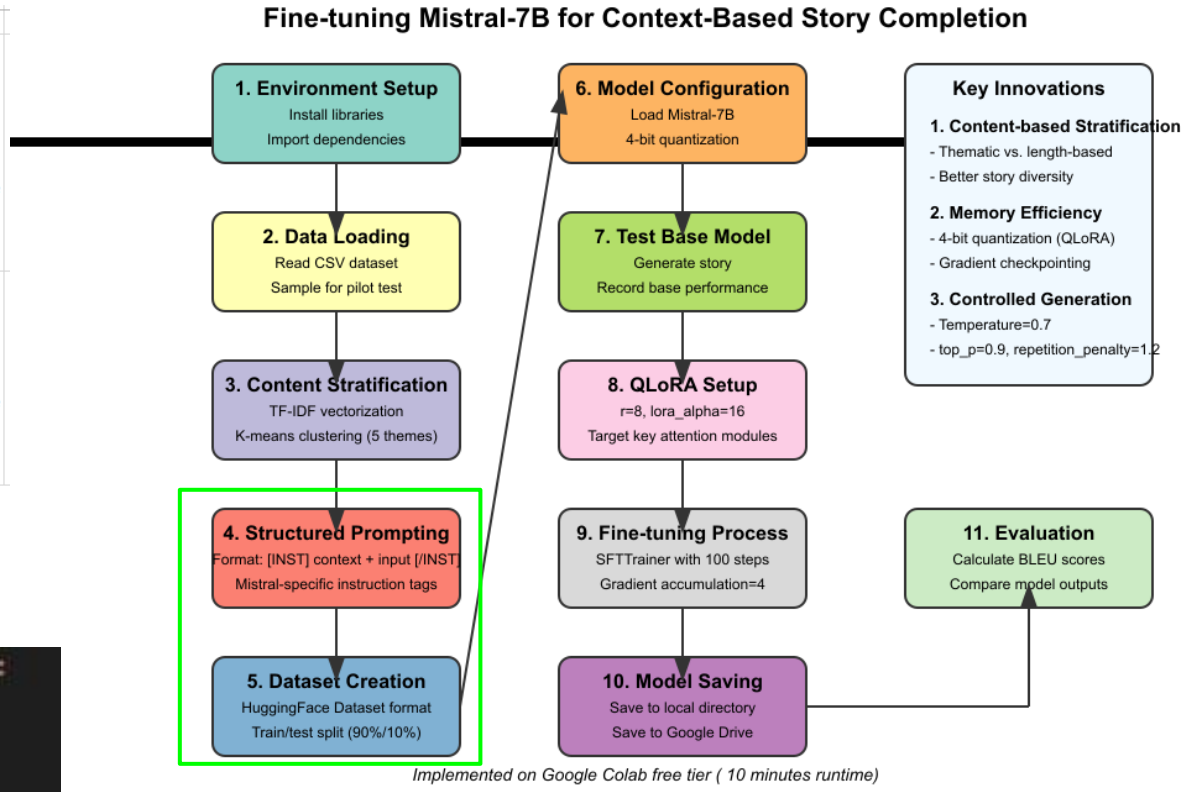


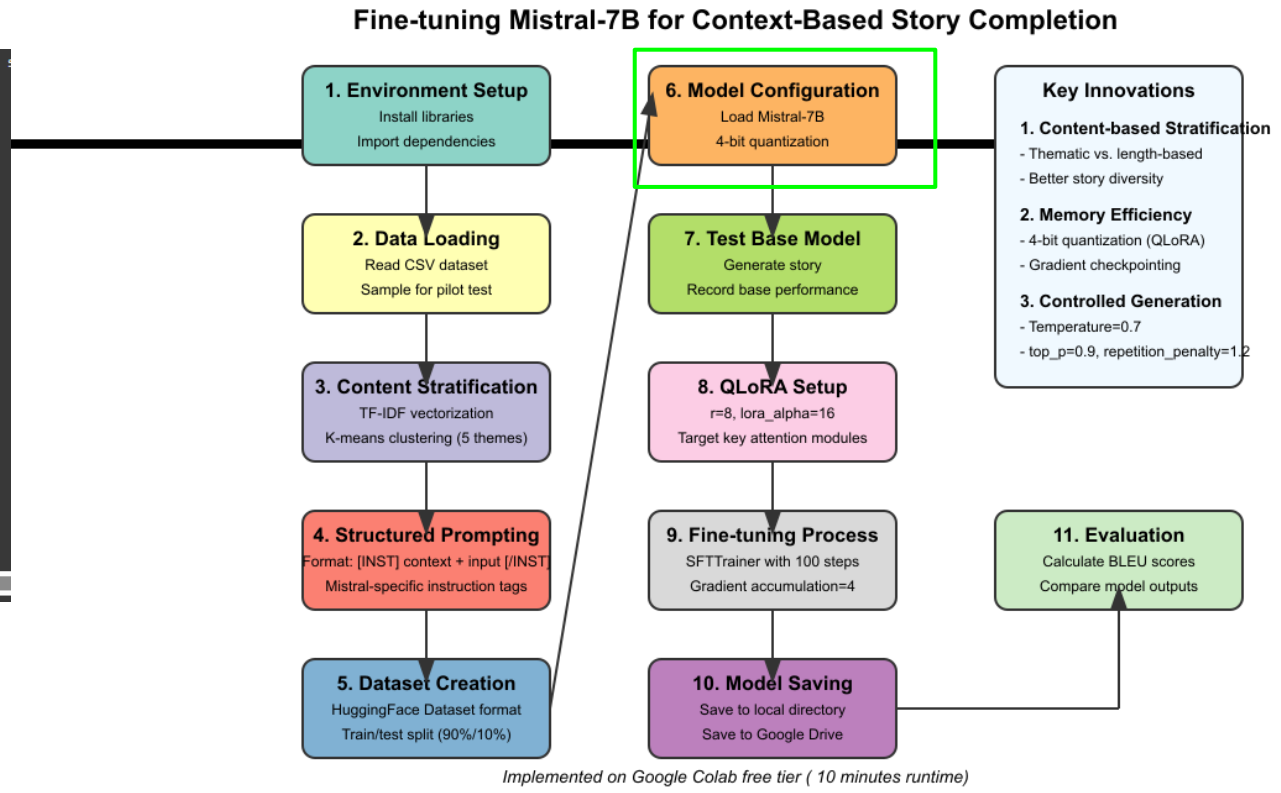

```
    prompt = f"""<s>[INST] Continue this story concisely (1-2 paragraphs):
Context: {row['context']}
Input: {row['input']} [/INST]"""

    target = row['output']
    return {
        "prompt": prompt,
        "target": target,
        "formatted_text": f"{prompt} {target}</s>"
    }
```

```
Sample formatted example:
<s>[INST] Continue this story concisely (1-2 paragraphs):
Context: John O'Keeffe, Professor of Philosophy at Reiser College, explains to Vincent D. van Dongen his discovery of a sphere two feet in diameter dated September 24, 1849. He says the sphere was returned to the university for study and that photogr
Input: Since then, he has been teaching philosophy [/INST] Since then, he has been teaching philosophy since then. Fisher explains how he came to create the sphere, explaining that he built models of the figures projected through a glass globe with d
```

```
The narrator notes that this is not the only problem: there are other problems as well. For example, one side of the sphere displays three hundred and sixty characters, whereas the other sides show only eighteen characters. In other words, it has no b
bout three hundred unique characters. At the end of the chapter, Dr. Karl Schwartz, head of the group, states that "This is the complete translation in idiomatic English.</s>
```

```
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
tokenizer_config.json: 100%  2.10k/2.10k [00:00<00:00, 228kB/s]
tokenizer.model: 100%  493k/493k [00:00<00:00, 11.1MB/s]
special_tokens_map.json: 100%  414/414 [00:00<00:00, 35.1kB/s]
tokenizer.json: 100%  1.80M/1.80M [00:00<00:00, 8.23MB/s]
config.json: 100%  571/571 [00:00<00:00, 37.5kB/s]
model.safetensors.index.json: 100%  25.1k/25.1k [00:00<00:00, 1.87MB/s]
Downloading shards: 100%  2/2 [01:26<00:00, 39.76s/it]
model-00001-of-00002.safetensors: 100%  9.94G/9.94G [01:04<00:00, 178MB/s]
model-00002-of-00002.safetensors: 100%  4.54G/4.54G [00:22<00:00, 246MB/s]
Loading checkpoint shards: 100%  2/2 [01:05<00:00, 30.37s/it]
generation_config.json: 100%  116/116 [00:00<00:00, 7.14kB/s]
```



4-bit Quantization

Testing with base model (before fine-tuning):

Prompt: **<s>[INST]** Continue this story concisely (1-2 paragraphs):

Context: Octavius Spink throws a bombshell at a meeting of cosmogonists at the University of Cincinnatus. He tells them that he intends to explore outer space, using hydrogen and oxygen from his "Hydroactive" capsule as fuel. They start out in about a foot of topsoil, then some hard rock and then more hard rock. Then another earthquake shakes the real estate of Department X6 near the Rockies that makes the Grand Canyon look like a kid just scraped a stick through some mud. Octavius plans to go seventy million miles into space by March 2022 without permission from the Board. The Euthanasia Society refuses to allow people to commit suicide in the year 2022 because they believe it will spill all the petroleum out of the earth all at once, so fast they couldn't refine it.

Input: Meanwhile, Gulflex and other oil companies protest **[/INST]**

Base model completion:

As news of Octavius Spink's audacious plan to venture into outer space spreads beyond the halls of academia, public outrage ensues. Gulflex and other major oil companies lead the charge against Spink, arguing that his unauthorized exploration could disrupt critical resources needed for their businesses. With global economies heavily reliant on fossil fuels, the potential consequences of such an endeavor are significant. Amidst mounting pressure from both scientific communities and industry leaders, the Board of the University of Cincinnatus is forced to intervene, demanding that Spink halts all preparations until proper authorization can be granted. Despite these obstacles, Spink remains resolute in his determination to push forward with the mission, raising questions about the boundaries of scientific progress and individual ambition.

```
Testing with base model (before fine-tuning):
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
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Input: Meanwhile, Gulflex and other oil companies protest [/INST]

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As news of Octavius Spink's audacious plan to venture into outer space spreads beyond the halls of academia, public outrage ensues. Gulflex and other major oil companies lead the charge against Spink, arguing that his unauthorized exploration could d
```
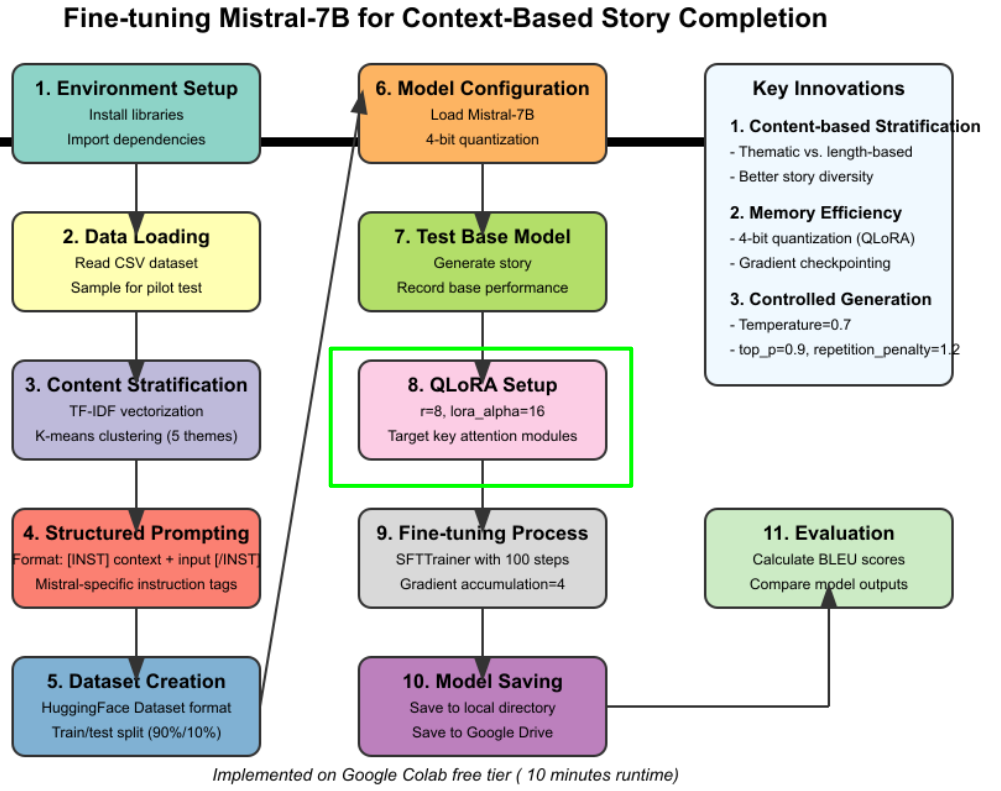
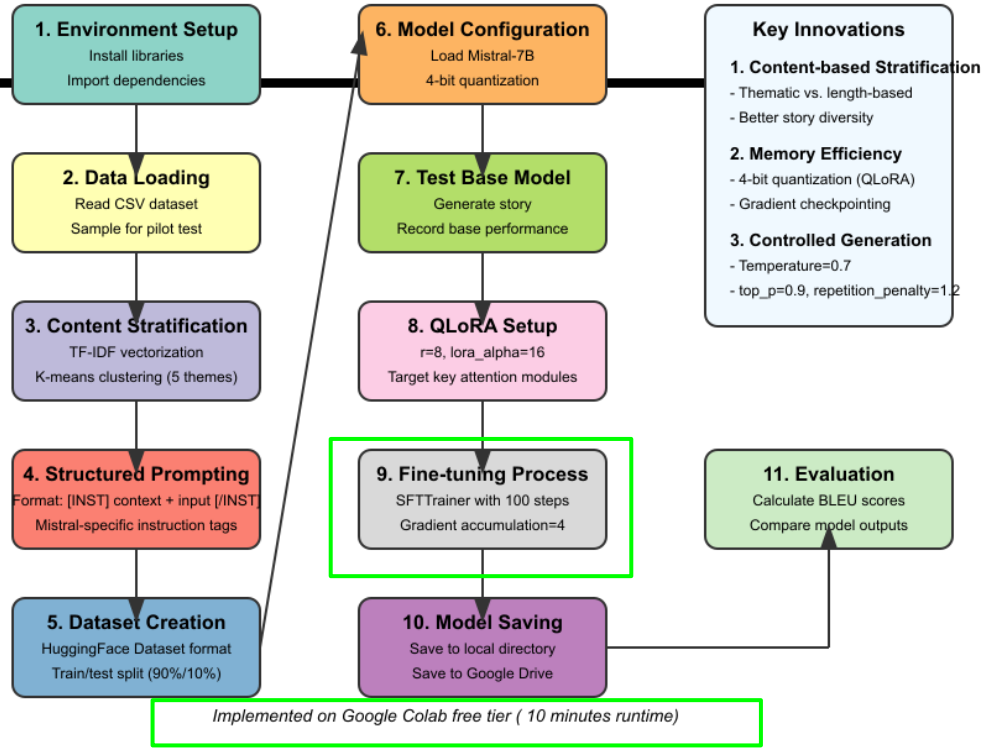

```python
peft_config = LoraConfig(
    r=8,                        # Lower rank (8 instead of 16) for memory efficiency
    lora_alpha=16,              # Scale parameter
    # Target key modules but fewer, for efficiency
    target_modules=["q_proj", "v_proj", "k_proj", "o_proj"],
    lora_dropout=0.1,           # Dropout for regularization
    bias="none",                # Don't apply LoRA to biases
    task_type="CAUSAL_LM",      # For causal language modeling
)

# Apply LoRA config to the model
model = get_peft_model(model, peft_config)
print("LoRA parameters added to model")

# Display trainable vs non-trainable parameters
model.print_trainable_parameters()
```

```
LoRA parameters added to model
trainable params: 6,815,744 || all params: 7,248,547,840 || trainable%: 0.0940290959023318
```



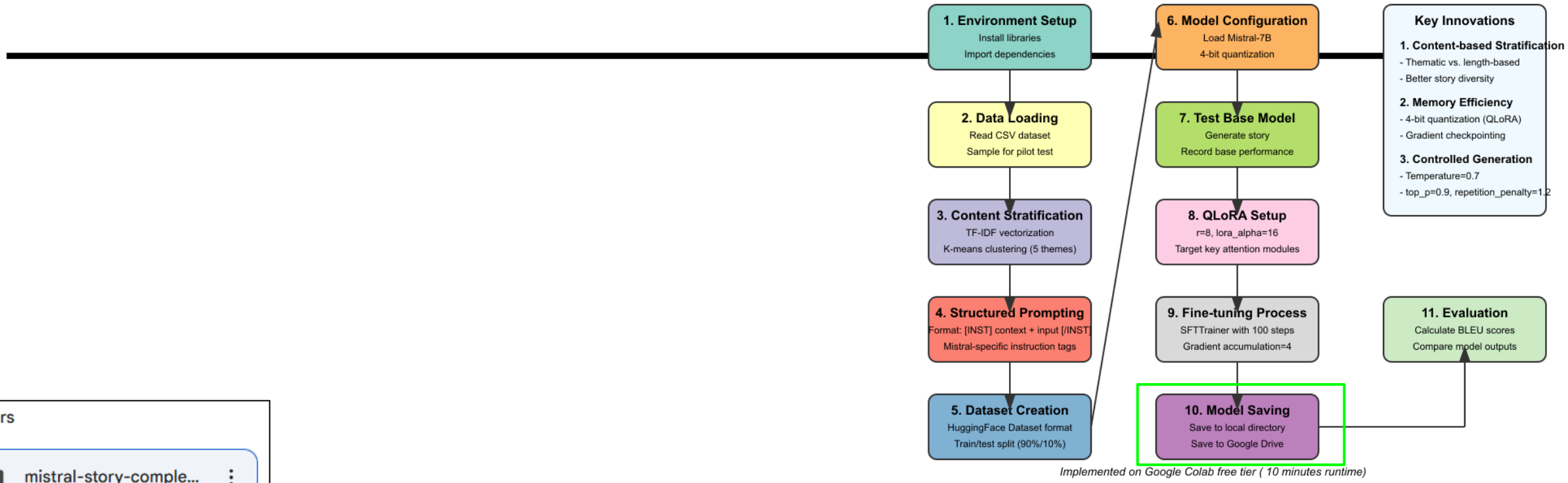
Fine-tuning Mistral-7B for Context-Based Story Completion

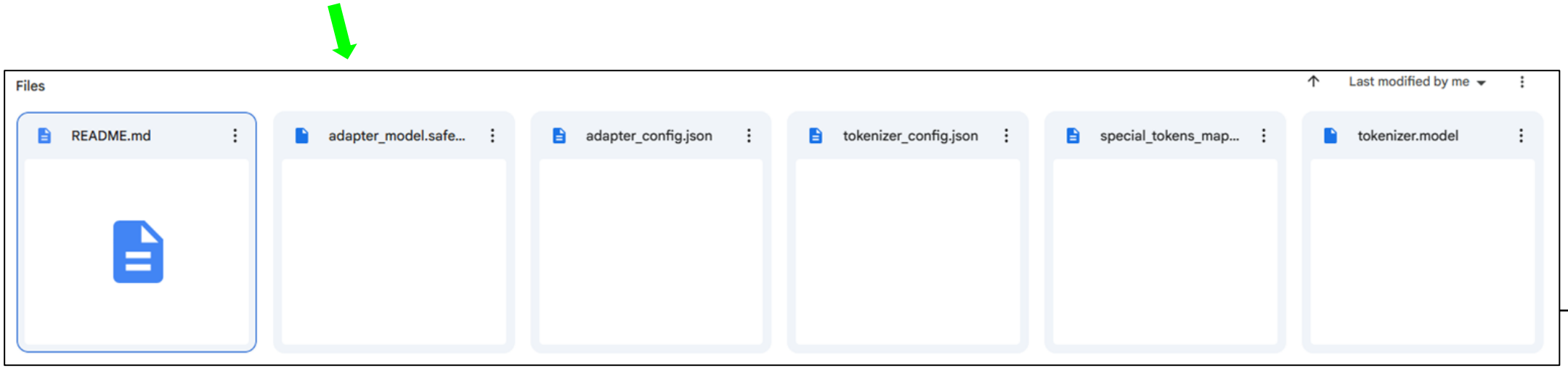



```
Starting fine-tuning...
`use_cache=True` is incompatible with gradient checkpointing. Setting `use_cache=False`...
[100/100 12:52, Epoch 3/4]
```

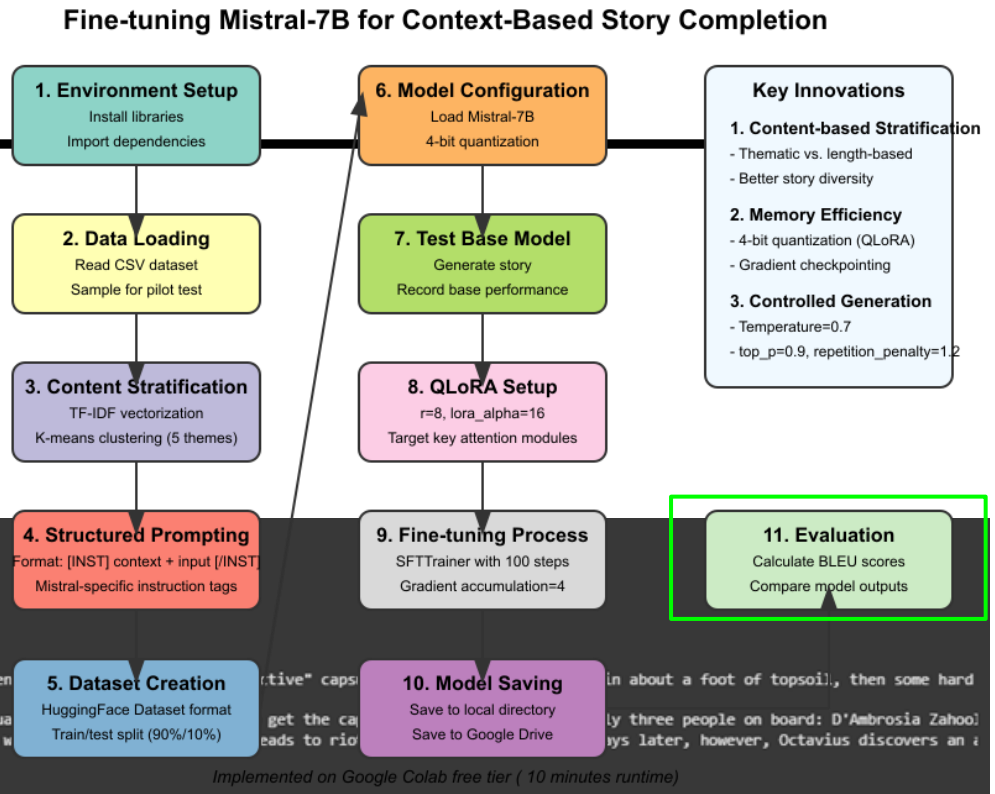
```
Training completed in 779.82 seconds (13.00 minutes)
```

Fine-tuning Mistral-7B for Context-Based Story Completion





RESULT:




```
Evaluating on validation set...
Average BLEU score on validation set: 0.0525

Sample evaluations:

Example 1:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: Octavius Spink throws a bombshell at a meeting of cosmogonists at the University of Cincinnatus. He tells them that he intends to explore outer space, using hydrogen
Input: Meanwhile, Gulflex and other oil companies protest [/INST]
Reference: Meanwhile, Gulflex and other oil companies protest to Number One as they say they would not refine it if they did not have enough hydrogen for the capsule. Eventua
Generated: Meanwhile, Gulflex and other oil companies protest against Spinks plan. They claim that if anyone tries to get rid of their jobs or their money, there will be no w
BLEU score: 0.0593

Example 2:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: Johnny Ludlow describes his life at Dyke Manor, the home of Squire Todhetley and his wife, Mrs. Ludlow. His step-mother is a wealthy landowner who lives in a good old-fashioned manner. He has two illegitimate children: a son named after his father, Johnny, and a daughter nam
Input: Her parents died as well. [/INST]
Reference: Her parents died as well. She goes on to be a very nice young lady. One day, they come upon a group of gypsies stealing children from the Sterlings' house. Hannah tries to run away, but Tod grabs her and tells her to stay where she is.
Generated: Her parents died as well. She was left alone on her hands until she married Sir Leicester Dedlock, an elderly baronet. After several months, however, he decided to leave London because he did not want to be seen again by any members of his family. One day while walking the
BLEU score: 0.0525

Example 3:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: "Now, then! What seems to be the trouble?" Wheatley had been palpitating for five days straight at the prospect of coming here. "I know it's silly," he says. "But I've been having a pain in my toe." "My dear fellow!" "Your toe is part of you. It doesn't just exist down there
Input: There's nothing to worry about except treating [/INST]
Reference: There's nothing to worry about except treating symptoms. Medieval quackery. Medicine has advanced a long way since his last checkup. And even treatment has its dangers.
Generated: There's nothing to worry about except treating it with kindness and consideration. This will give you something else to think about while your toe gets better. When you have finished speaking to Mr. Holmes, please return to your seat and take care not to touch anything tha
BLEU score: 0.0389

Example 4:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: John O'Keeffe, Professor of Philosophy at Reiser College, explains to Vincent D. van Dongen his discovery of a sphere two feet in diameter dated September 24, 1849. He says the sphere was returned to the university for study and that photographs of the characters projected
Input: The narrator notes that this is not [/INST]
Reference: The narrator notes that this is not the only problem: there are other problems as well. For example, one side of the sphere displays three hundred and sixty characters, whereas the other sides show only eighteen characters. In other words, it has no bearing on the actual
Generated: The narrator notes that this is not exactly what it seems to be; rather, he points out some oddities about the device. For example, when Van Dongen looks into one of the tubes on the surface of the glass ball , there are no markings or anything else visible except for a s
BLEU score: 0.0653

Example 5:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: nan
Input: The novel opens with a description of [/INST]
Reference: The novel opens with a description of Crescent Bay. The beach is uninhabited, and there is nothing to indicate where the road ends or where the bungalows begin. A flock of sheep are huddled together in a sheltered corner of the sandbar. An old shepherd comes into the scene
Generated: The novel opens with a description of the main character, Tom Brinston. He is described as being "a little older than fifty," having "grayish" hair and wearing spectacles. We also learn that he has lived in his current home for twenty years. In other words, he's been livi
The narrator then tells us about another person who lives on this same street--an old man named John Baxter. Unlike most people, however, Tom doesn't mind sharing his house with someone else. Instead, he spends most of his time trying to figure out what exactly happened to him back
BLEU score: 0.0463
```

Evaluating on validation set...
Average BLEU score on validation set: **0.0525**

Sample evaluations:

Even though not very similar to reference, Good enough for given context & input

Example 1:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: Octavius Spink throws a bombshell at a meeting of cosmogonists at the University of Cincinnatus. He tells them that he intends to explore outer space, using hydrogen and oxygen from his "Hydroactive" capsule as fuel. They start out in about a foot of topsoil, then some hard rock and then more hard rock. Then another earthquake shakes the real estate of Department X6 near the Rockies that makes the Grand Canyon look like a kid just scraped a stick through some mud. Octavius plans to go seventy million miles into space by March 2022 without permission from the Board. The Euthanasia Society refuses to allow people to commit suicide in the year 2022 because they believe it will spill all the petroleum out of the earth all at once, so fast they couldn't refine it.
Input: Meanwhile, Gulflex and other oil companies protest [/INST]
Reference: Meanwhile, Gulflex and other oil companies protest to Number One as they say they would not refine it if they did not have enough hydrogen for the capsule. Eventually, however, they manage to get the capsule up and running with only three people on board: D'Ambrosia Zahooli, D'Amrosia Zmorro, and Axitope Wurpz. All three crew members are aboard the capsule when an earthquake knocks the entire population of the Magnificent Mole over the edge of the Earth's crust. It is Friday morning, October 21st.
Generated: Meanwhile, Gulflex and other oil companies protest against Spinks plan. They claim that if anyone tries to get rid of their jobs or their money, there will be no way back for humanity. This leads to riots in New York City. A few days later, however, Octavius discovers an asteroid belt around the sun called "Octavia." He decides to name it after himself since no one else has claimed any part of space yet. When he gets home, though, his wife asks him why he doesn't just stay on Earth instead of going off into space . She says that she wants him to come back when everything is settled down again. After all, what good would it do for her to live on Mars or Venus while everyone else was still stuck here? In response, Octavius explains how much money he owes in loans and debts. His wife argues that he should have made enough money by now; but he points out that she
BLEU score: 0.0593

Example 2:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: Johnny Ludlow describes his life at Dyke Manor, the home of Squire Todhetley and his wife, Mrs. Ludlow. His step-mother is a wealthy landowner who lives in a good old-fashioned manner. He has two illegitimate children: a son named after his father, Johnny, and a daughter named after her mother, Eliza. During the holidays, Tod and Hugh go to live at Bleak House, owned by William Ludlow's father, Mr. Todheley. They are there for a term of years. The Squire dies that same year. Lena comes to live with them; she is orphaned and widowed.
Input: Her parents died as well. [/INST]
Reference: Her parents died as well. She goes on to be a very nice young lady. One day, they come upon a group of gypsies stealing children from the Sterlings' house. Hannah tries to run away, but Tod grabs her and tells her to stay where she is.
Generated: Her parents died as well. She was left alone on her hands until she married Sir Leicester Dedlock, an elderly baronet. After several months, however, he decided to leave London because he did not want to be seen again by any members of his family. One day while walking through Hyde Park, Lena saw him talking to someone else. When they approached each other, she heard his voice asking whether anyone would like to join him at the party tonight. As she walked back towards the house, she realized that it must have been one of those dark days when the sun had shone out so brightly. It wasn't long before everyone started calling her Lady Dedlock. At first, no one knew what to call her except "lady"--but soon enough people began referring to her simply as "Lady." That summer evening, Lena went down into the garden to pick some flowers from the lawn near where she lived. While doing so
BLEU score: 0.0525

Example 3:
Prompt: <s>[INST] Continue this story concisely (1-2 paragraphs):
Context: John O'Keeffe, Professor of Philosophy at Reiser College, explains to Vincent D. van Dongen his discovery of a sphere two feet in diameter dated September 24, 1849. He says the sphere was returned to the university for study and that photographs of the characters projected by the sphere caused him to believe they had origin from an unknown culture.
Input: The narrator notes that this is not [/INST]
Reference: The narrator notes that this is not the only problem: there are other problems as well. For example, one side of the sphere displays three hundred and sixty characters, whereas the other sides show only eighteen characters. In other words, it has no bearing on the actual history of the world or its inhabitants. Van Dongen tells Fisher that he first encountered the sphere eight months ago when he was an undergraduate.
Generated: The narrator notes that this is not exactly what it seems to be; rather, he points out some oddities about the device. For example, when Van Dongen looks into one of the tubes on the surface of the glass ball , there are no markings or anything else visible except for a series of black spots scattered across its face like stars on a night sky . When Van Dongen tries to take another look inside the tube , however, nothing happens . Instead, a small sphere appears in front of the professor 's head . It floats around and then disappears again after a few minutes. This time, though, there are still more spots than before! Now we see them moving rapidly over each other as if they were trying to get closer together. After a while, however, they all start swirling together until finally everything goes dark. That's right--the whole thing just disappeared! Well, maybe not entirely--there's always been something strange going
BLEU score: 0.0653

Human Evaluation:

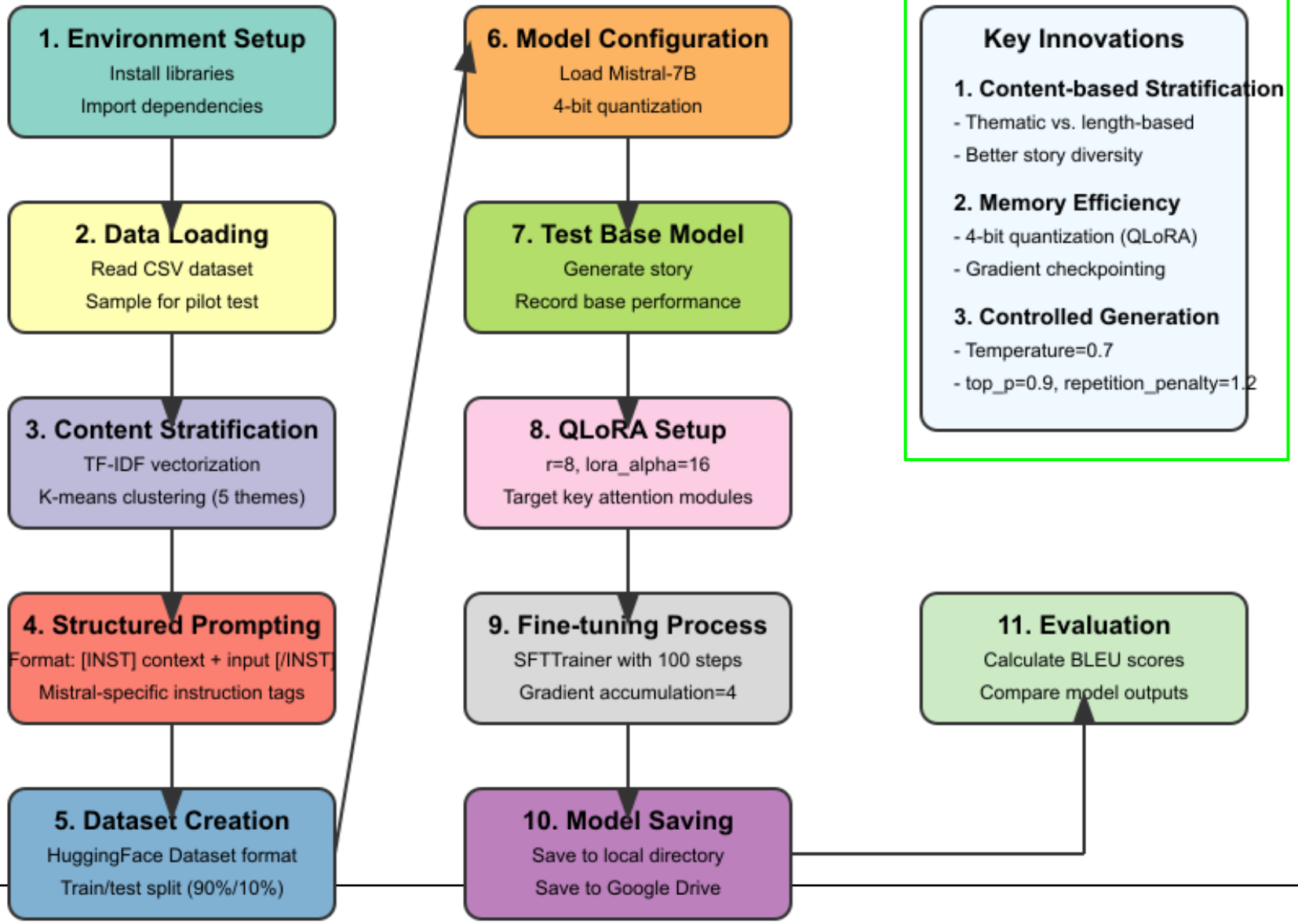
(Since BLEU doesn't work on data that model has never seen before)

Context	Input	Generated	Linguistic Coherence	Contextual Coherence	Creativity
Octavius Spink throw	Meanwhile, Gulflex a	Meanwhile, Gulf	9	8	9
Johnny Ludlow descr	Her parents died as	Her parents died	9	9	8
"Now, then! What see	There's nothing to w	There's nothing	8	8	7
John O'Keeffe, Profes	The narrator notes t	The narrator not	8	9	10

Averages:

- Linguistic Coherence - **9.10**
- Contextual Coherence - **8.23**
- Creativity - **8.81**

Fine-tuning Mistral-7B for Context-Based Story Completion



Implemented on Google Colab free tier (10 minutes runtime)

CHALLENGES AND LIMITATIONS…

- Challenges –
  - **Resource Constraints**: Implementation across different model architectures required significant optimization to operate within consumer-grade hardware limitations.
  - **Data Quality & Balance**: Ensuring training data had sufficient diversity and quality for storytelling without introducing biases or repetitive patterns.
- Limitations –
  - **Model Size Restrictions**: Current implementation limited to models of 3-7B parameters due to memory constraints, excluding larger models that might yield better performance.
  - **Generation Length**: Limited context window restricts the ability to maintain coherence in longer narrative sequences.

Thank you.